



10264 - The First Stars or the Last Stars: Confirmation of a Robust Candidate $z \sim 30$ Galaxy

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	z30	NIRSpec MultiObject Spectroscopy	(1) goodsn_candidates_weights

ABSTRACT

Following early JWST discoveries of an overabundance of $z \sim 10$ -15 galaxies, candidates at $z \sim 15$ -25 have recently been discovered. How early did the universe form galaxies? We have searched for candidates at even higher redshifts, finding five F356W-dropout galaxies. While exhibiting colors consistent with $z > 25$, they are also be consistent with an ultra-cool (325 K) low-metallicity halo Y dwarf, or a faint accreting supermassive black hole at $z > 9$.

JWST NIRSpec prism spectroscopy can robustly discern between these scenarios. Confirmation of a $z > 25$ galaxy would rewrite our understanding of the earliest phases of galaxy formation just 100 Myr after the Big Bang. Discovery of a metal-poor room-temperature substellar object would place critical constraints on the formation of the lowest mass substellar objects early in cosmic time, while confirmation of an accreting black hole at $z > 9$ would further push our massive black hole horizons.

We propose to observe the most robust candidate, in the GOODS-N field, where multiple NIRCам epochs, NIRCам grism, and MIRI data rule out other scenarios (e.g., strong lines, dusty star-forming galaxies, or lower-redshift SNe), and we can observe two $z \sim 30$ candidates with the MSA.

In 16 hr we can discern a $z \sim 30$ Ly-alpha break from either a Y-dwarf or a LRD at > 5 -sigma significance. If the true origin is a halo Y-dwarf, these data will be able to robustly constrain the metallicity, while in the LRD scenario detection of broad H-beta will constrain the black hole mass.

Whether a "first star", a metal poor "last star", or a high- z black hole, this object's origin will provide critical insights into star formation in the early or late universe.

OBSERVING DESCRIPTION

We will obtain deep, 16 hour NIRSpec MSA PRISM spectra of two robust $z \sim 30$ candidate galaxies in the GOODS-N field. We will use these observations to definitively differentiate between a) a galaxy at $z \sim 30$, b) a $z \sim 9-12$ little red dot (LRD), or c) a substellar low-metallicity Y-dwarf in the Milky Way. To maximize the legacy value of these deep PRISM observations, we will also fill the MSA with an additional 200 nearby objects including 65 at $6 < z < 9$, and 6 at $z > 9$.

In order to differentiate between the three afore-mentioned scenarios, we require 16 hours of PRISM exposure to measure the shape of the proposed Lyman-alpha break at 5-sigma significance, and differentiate between different continuum slopes at a 3-sigma significance. We will use three shutter slitlets, a three point nodding pattern, and the NRSIRS2 readout mode with 22 groups per integration and two integrations per exposure. We will repeat this pattern six times to reach our required 15 hours of exposure time.

Proposal 10264 - Targets - The First Stars or the Last Stars: Confirmation of a Robust Candidate z~30 Galaxy

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	goodsn_candidates_weights	RA: 12 36 59.1672 (189.2465300d) Dec: +62 11 35.66 (62.19324d) Equinox: J2000		
	Comments: Description=[]				

Proposal 10264 - Observation 1 - The First Stars or the Last Stars: Confirmation of a Robust Candidate z~30 Galaxy

Observation	Proposal 10264, Observation 1: z30											Wed Apr 29 04:00:31 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec MultiObject Spectroscopy												
Diagnostics	(z30 (Obs 1)) Warning (Form): Config c1 (#1) has 2 filler slits affected by failed closed shutters.											
	(z30 (Obs 1)) Warning (Form): Config c1 (#2) has 2 filler slits affected by failed closed shutters.											
	(z30 (Obs 1)) Warning (Form): Config c1 (#3) has 2 filler slits affected by failed closed shutters.											
	(z30 (Obs 1)) Warning (Form): Config c1 (#4) has 2 filler slits affected by failed closed shutters.											
	(z30 (Obs 1)) Warning (Form): Config c1 (#5) has 2 filler slits affected by failed closed shutters.											
	(z30 (Obs 1)) Warning (Form): Config c1 (#6) has 2 filler slits affected by failed closed shutters.											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	goodsn_candidates_weights	RA: 12 36 59.1672 (189.2465300d) Dec: +62 11 35.66 (62.19324d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold				
	MSATA	false	No	MSA Center	Primary (49 sources)	Filler (19660 sources)	jwst-nirspec-prism	1.5				
Reference Stars												

Proposal 10264 - Observation 1 - The First Stars or the Last Stars: Confirmation of a Robust Candidate z~30 Galaxy

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	189.24575329166 666 Degrees 62.191746944444 44 Degrees	74.999309878575 45			3	6	9716.201
	2	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	189.24575329166 666 Degrees 62.191746944444 44 Degrees	74.999309878575 45			3	6	9716.201
	3	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	189.24575329166 666 Degrees 62.191746944444 44 Degrees	74.999309878575 45			3	6	9716.201
	4	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	189.24575329166 666 Degrees 62.191746944444 44 Degrees	74.999309878575 45			3	6	9716.201
	5	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	189.24575329166 666 Degrees 62.191746944444 44 Degrees	74.999309878575 45			3	6	9716.201
	6	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	189.24575329166 666 Degrees 62.191746944444 44 Degrees	74.999309878575 45			3	6	9716.201
Special Requirements	Between Dates 16-DEC-2026:00:00:00 and 10-APR-2027:00:00:00 MSA Planned Aperture PA 75.0000 to 75.0000 Degrees (V3 296.4254303 to 296.4254303)										